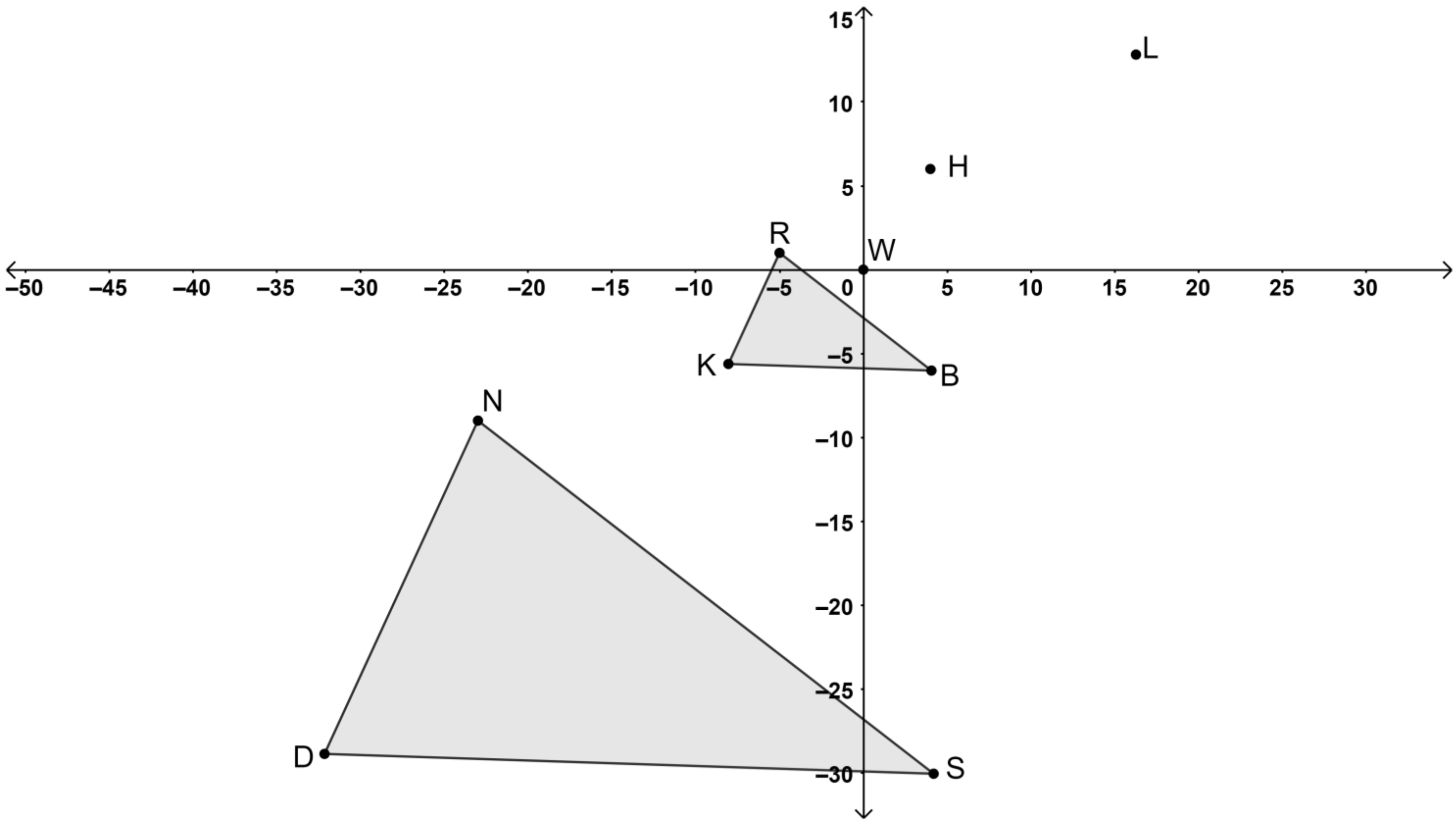
Geometry
Unit 6 Assessment Part 1

Name______________________
Date__________________
Period__________

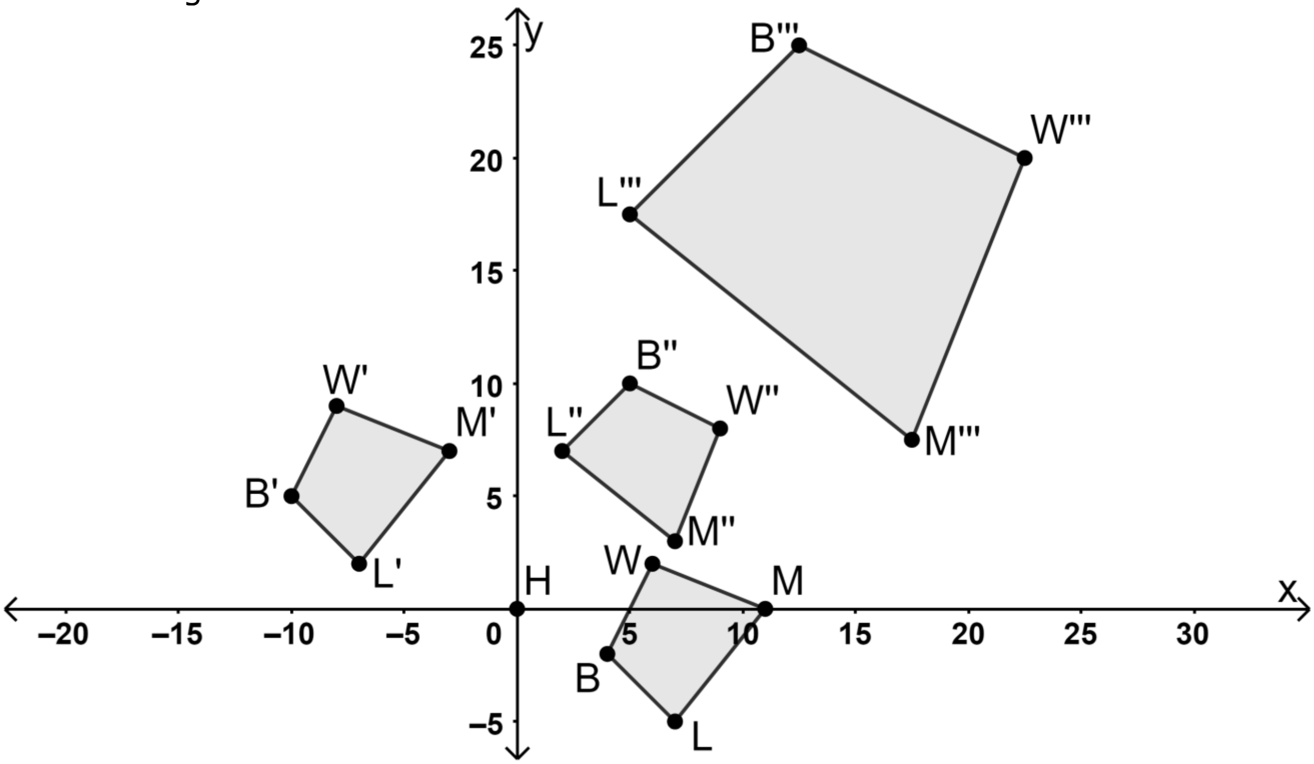
1. Triangles DNS and KRB are shown on the coordinate grid.



Complete the statement.

Triangle
○ DNS
○ KRB
is the result of dilating triangle
○ DNS
○ KRB
using a scale factor of 3 centered at
○ H.
○ L.
○ W.

2. Quadrilaterals $BLMW$, $B'L'M'W'$, $B''L''M''W''$, and $B'''L'''M'''W'''$ are shown on the coordinate grid.



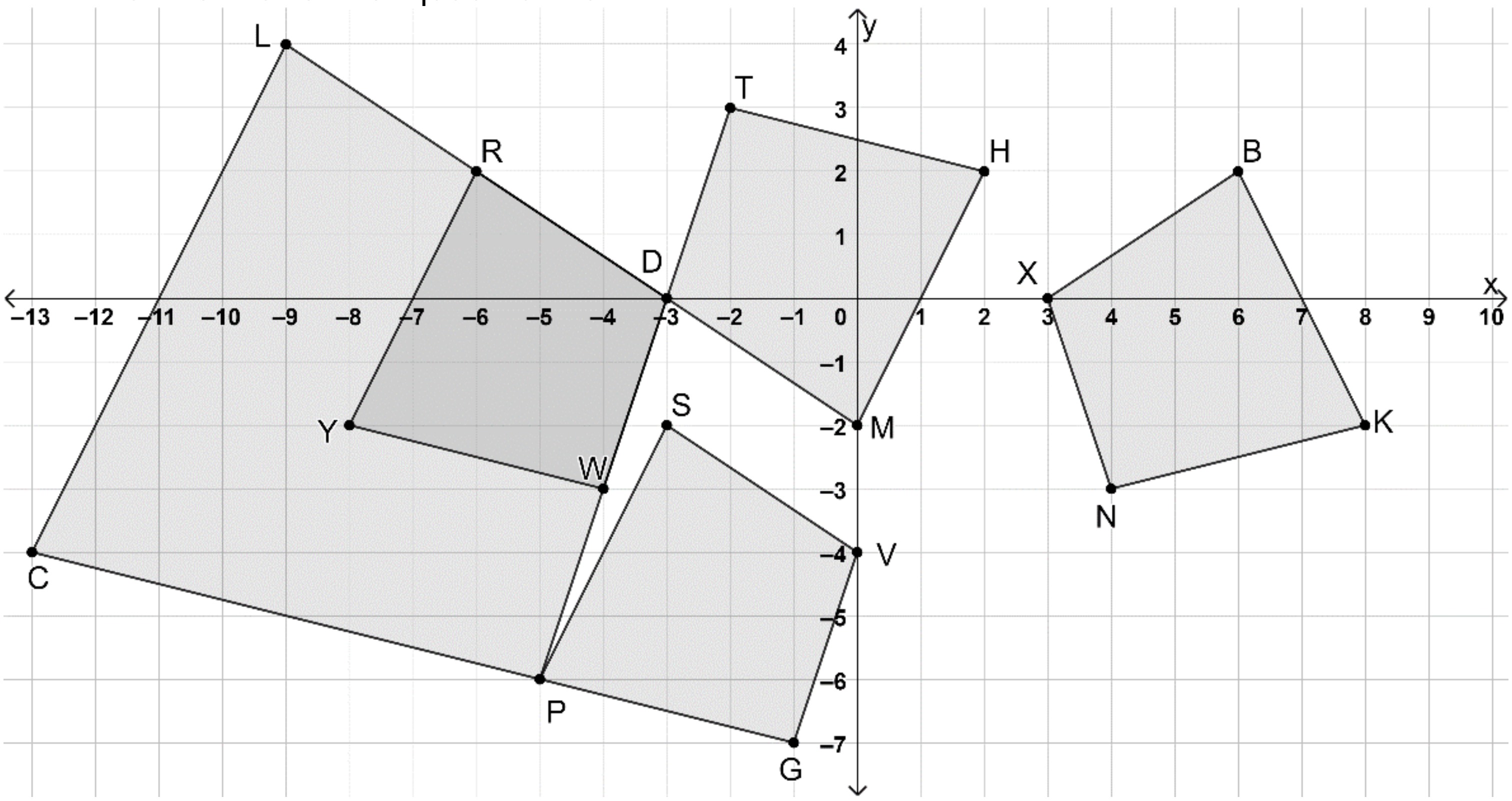
The coordinates of the preimage and each transformation are shown.

Preimage		Transformation 1		Transformation 2		Transformation 3	
B	$(4,-2)$	B'	$(-10,5)$	B''	$(5,10)$	B'''	$(12.5,25)$
L	$(7,-5)$	L'	$(-7,2)$	L''	$(2,7)$	L'''	$(5,17.5)$
M	$(11,0)$	M'	$(-3,7)$	M''	$(7,3)$	M'''	$(17.5,7.5)$
W	$(6,2)$	W'	$(-8,9)$	W''	$(9,8)$	W'''	$(22.5,20)$

Complete the table by describing each transformation and giving the algebraic description using coordinates.

	Written Description	**Algebraic Description**
Transformation 1	Translation left 14, up 7	$(x,y) \rightarrow (x-14, y+7)$
Transformation 2	Rotation $90°$ clockwise about the origin	$(x,y) \rightarrow (y,-x)$
Transformation 3	Dilation of 2.5 centered at the origin	$(x,y) \rightarrow (2.5x, 2.5y)$

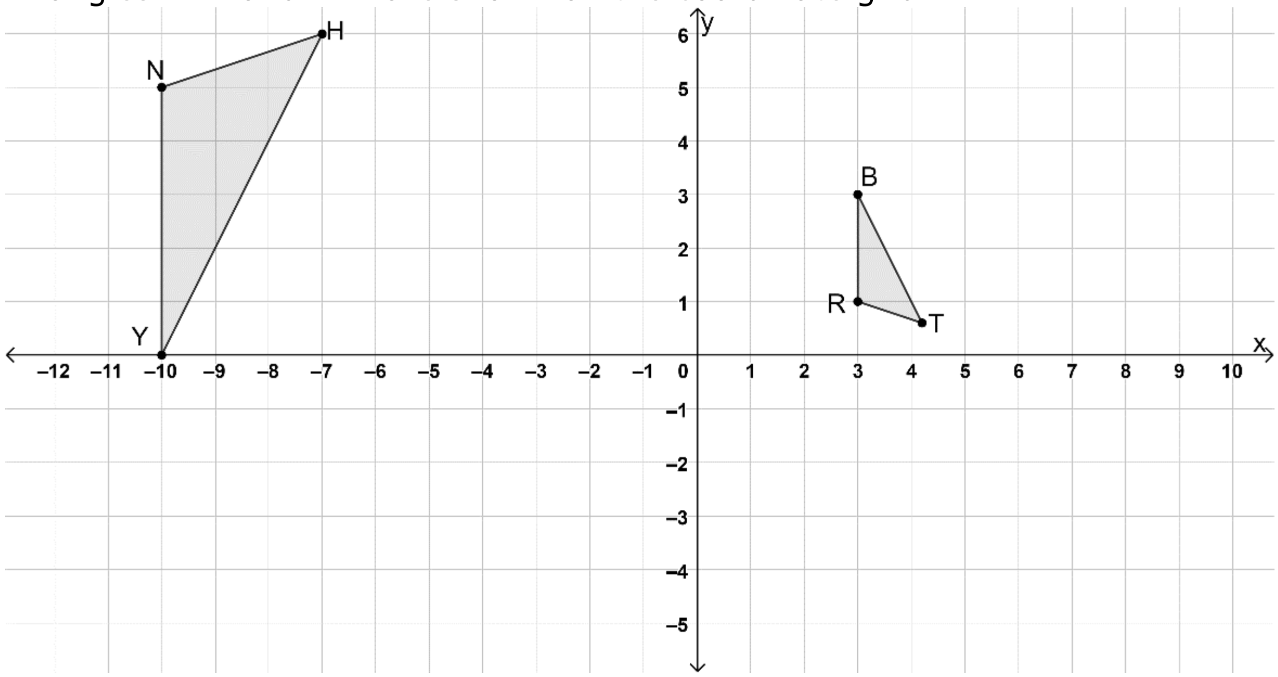
3. Quadrilaterals $RDWY$, $LDPC$, $TDMH$, $BXNK$, and $SVGP$ are shown on the coordinate grid, where quadrilaterals $LDPC$, $TDMH$, $BXNK$, and $SVGP$ are all transformations of quadrilateral $RDWY$.



Match each description with the quadrilateral that is the result of the transformation of quadrilateral $RDWY$. If none of the quadrilaterals match, then choose "none".

Transformation of quadrilateral $RDWY$	$LDPC$	$TDMH$	$BXNK$	$SVGP$	**none**
translation down 2	☐	☐	☐	☐	☐
translation right 6	☐	☐	☐	☐	☐
translation right 3 down 4	☐	☐	☐	☐	☐
dilation by a factor of 2 centered at point D	☐	☐	☐	☐	☐
dilation by a factor of $\frac{1}{2}$ centered at the origin	☐	☐	☐	☐	☐
reflection across y −axis	☐	☐	☐	☐	☐
reflection across $x = -3$	☐	☐	☐	☐	☐
rotation 180° about point D	☐	☐	☐	☐	☐
rotation 180° about the origin	☐	☐	☐	☐	☐

4. Triangles NHY and RBT are shown on the coordinate grid.



a. Select all of the sequences of transformations that will result in the triangles being mapped onto each other.

- ☐ ΔBRT maps to ΔYNH using $(x, y) \to (-y, x)$ then $(x, y) \to (x - 3, y - 4)$ then $(x, y) \to (2x, 2y)$
- ☐ ΔBRT maps to ΔYNH using $(x, y) \to (x - 7, y - 3)$ then $(x, y) \to (x, -y)$ then $(x, y) \to (2.5x, 2.5y)$
- ☐ ΔBRT maps to ΔYNH using $(x, y) \to (-y, -x)$ then $(x, y) \to (5x, 5y)$ then $(x, y) \to (x, -y)$
- ☐ ΔYNH maps to ΔBRT using $(x, y) \to (x + 6, y)$ then $(x, y) \to (x, -y)$ then $(x, y) \to (0.5x, 0.5y)$
- ☐ ΔYNH maps to ΔBRT using $(x, y) \to (0.4x, 0.4y)$ then $(x, y) \to (x, -y)$ then $(x, y) \to x + 7, y + 3)$

b. Using one of the mappings selected in part a, justify that triangles BRT and YNH are similar.

The mapping of triangle BRT to triangle YNH includes a dilation, so therefore the triangles are similar.

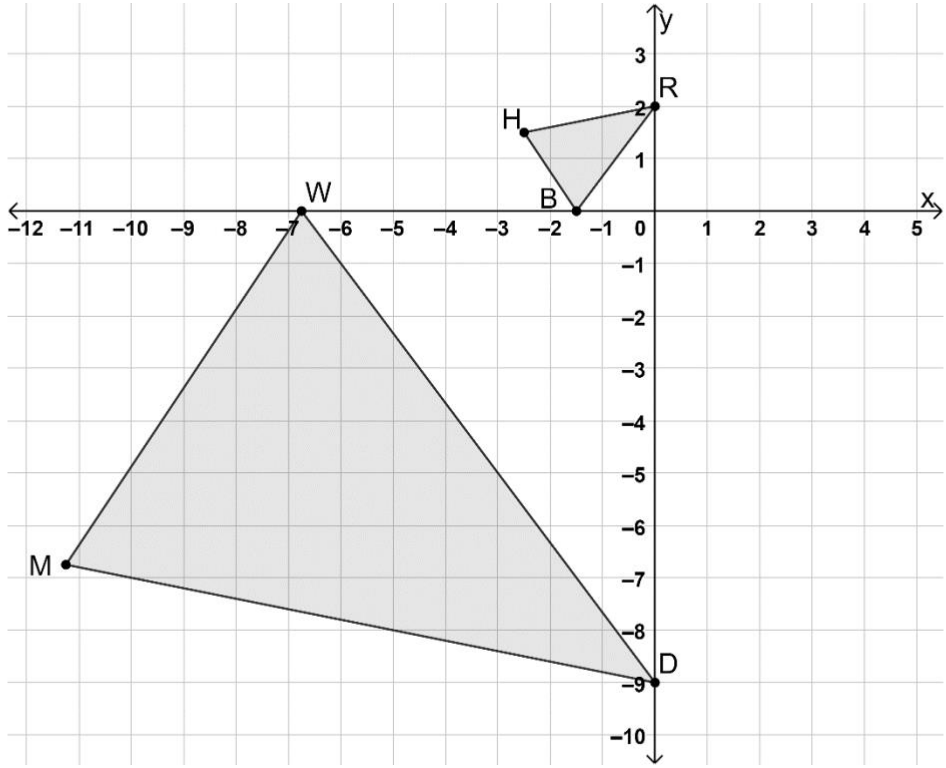
OR

The mapping of triangle YNH to triangle BRT includes a dilation, so therefore the triangles are similar.

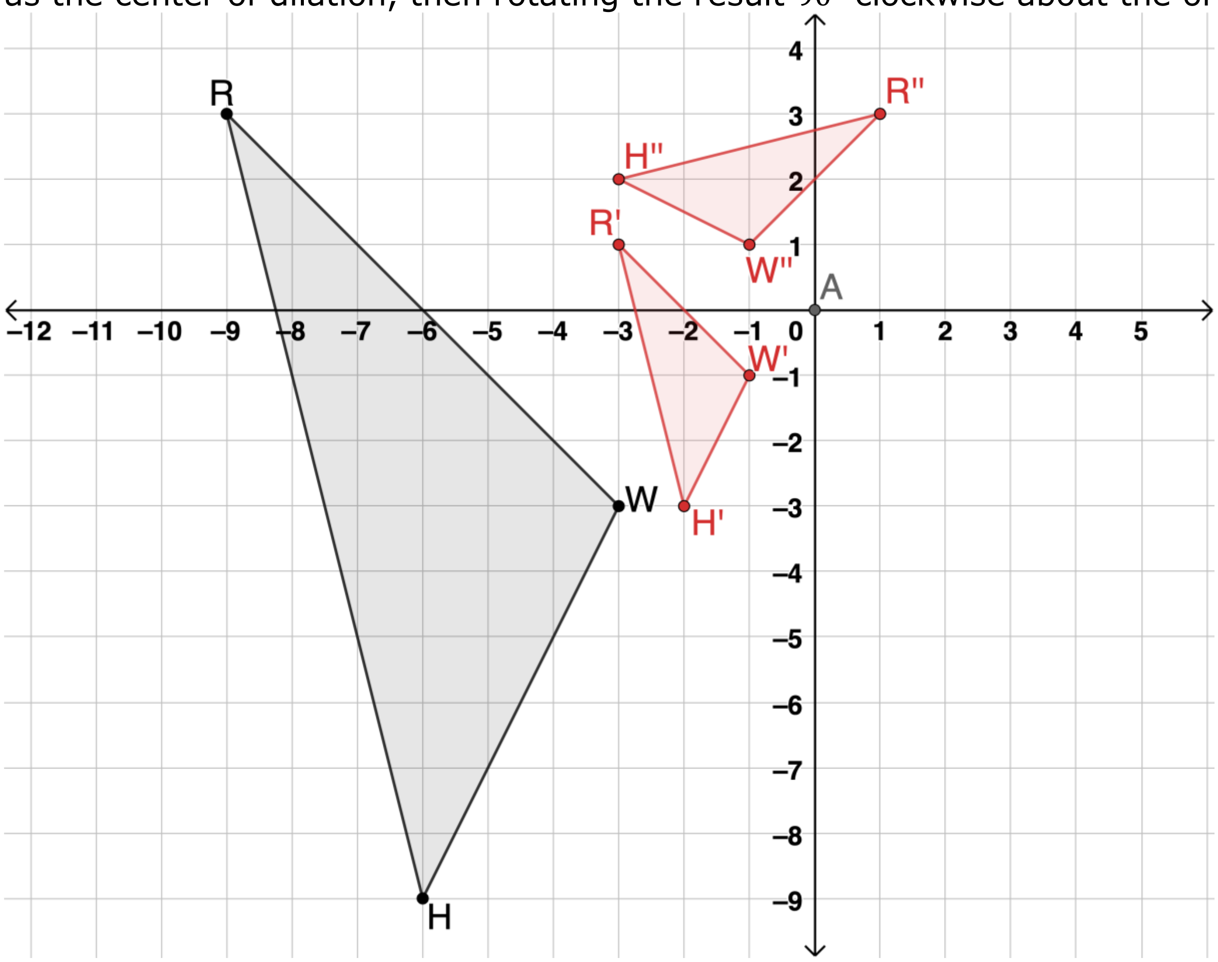
5. Triangles RHB and DMW are shown on the coordinate grid.

Write a sequence of transformations that will map ΔRHB onto ΔDMW.

Reflect ΔRHB across the x-axis, then dilate using a scale of 4.5 centered at the origin.

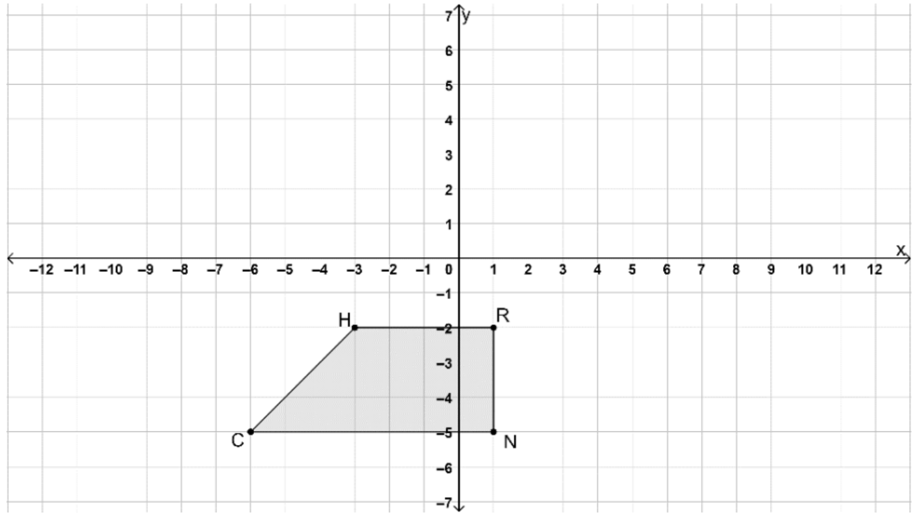


6. Triangle RWH is shown on the coordinate grid. On the same coordinate plane, draw $\Delta R''W''H''$, which is the result of dilating ΔRWH using a scale factor of $\frac{1}{3}$ with the origin as the center of dilation, then rotating the result $90°$ clockwise about the origin.



7. Quadrilateral $HRNC$ is shown on the coordinate grid.

Match each sequence of transformations of quadrilateral $HRNC$ with the characteristic(s) of $H''R''N''C''$, the result of the sequence of transformations.



Sequence of Transformations	**Preserves Distance**	**Preserves Angle Measure**	**$H''R''N''C''$ is similar but not congruent to $HRNC$**
$(x, y) \rightarrow (-y, x)$ then $(x, y) \rightarrow (3.5x, 3.5y)$	☐	☐	☐
$(x, y) \rightarrow \left(\frac{2}{3}x, \frac{2}{3}y\right)$ then $(x, y) \rightarrow (x, -y)$	☐	☐	☐
$(x, y) \rightarrow (x - 1, y + 5)$ then $(x, y) \rightarrow (x, y)$	☐	☐	☐
$(x, y) \rightarrow (-x, -y)$ then $(x, y) \rightarrow (x + 6, y - 2)$	☐	☐	☐